

Porcelain Veneers

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The porcelain veneer has gained wide acceptance in recent years as a primary restoration in esthetic dentistry. Since its introduction in the early 1980s, it has undergone an evolution in both techniques and materials. A significant number of long-term clinical studies confirm the excellent durability of the porcelain veneer restoration.[1–7]

Indications and Limitations

Because of their conservative preparation and beautiful esthetics, porcelain veneers have become the treatment of choice for restoration of the anterior dentition.[8] Porcelain veneers may be used to modify a tooth's color, shape, length, and/or alignment; to close space; and to restore fractured and endodontically treated teeth. The patient should be informed, however, of the possible morbidity associated with a specific indication as well as the generally accepted limitations. Informed consent should include, but not be limited to, the following possible complications: *(1)* postoperative sensitivity, *(2)* marginal discoloration, *(3)* fracture, *(4)* debonding, and *(5)* wear of opposing teeth.

Treatment Planning

The patient's self-image must be considered during the initial patient interview.[9] A key element in the diagnostic phase is a clarification of the patient's expectations. If it can be determined initially that the patient's expectations are unrealistic, future disappointment may be avoided. Esthetic failures are as large a percentage of failure of veneers as technical failures.[5] A systematic approach to assessing the face and smile is every bit as important as preparation design and adhesive choice.

Assessment of the face

When a treatment plan is being developed that includes restoration of teeth in the esthetic zone, attention must be directed not only to the shape and color of the teeth but also to the shape of the face, the lips, the maxillary and mandibular lip lines, and the skin color. The teeth can be used to accentuate a positive feature or de-emphasize a negative feature. For example, a patient with a narrow face may desire longer and narrower teeth to emphasize the facial shape or shorter, rounded teeth to soften the narrowness of the face.

It is also important to evaluate skin color, especially if there is a possibility that it will change over time. For example, if porcelain veneers are being planned for a Caucasian patient with a dark tan, a determination must be made regarding the longevity of the tan prior to shade selection. If the dark tan is transient and skin color will revert to a lighter tone, this will significantly affect the decision on the color of the porcelain veneers. Veneers that appear to be bright and high in value against the tanned skin will look more yellow and lower in value as the skin tone becomes lighter. All of these parameters must be consciously considered during the diagnostic phase if consistently excellent results are to be obtained.

Assessment of the smile

After the facial features have been considered, attention must be directed to the smile and its components. During the initial interview, the dentist should pay close attention to the overall appearance of the patient's mouth as he or she speaks and is in repose. The dentist should note the maxillary incisal edge position in relation to the lower lip, the relationship of the maxillary incisal plane to the horizon, the amount of gingival display during smiling and speaking, the relationship of the anterior and posterior segments, and the overall quality of the smile. If the interpupillary line is parallel to the horizon, it may be used in evaluation of the

incisal plane. The dentist pulls the upper lip parallel to the interpupillary line and then uses the lip to evaluate the incisal plane for a cant (Fig 17-1). The acceptability of the incisal plane must be determined during the diagnostic phase, or discrepancies may be incorporated into the definitive restorations. (See chapter 3 for a more complete discussion of esthetic diagnosis.)

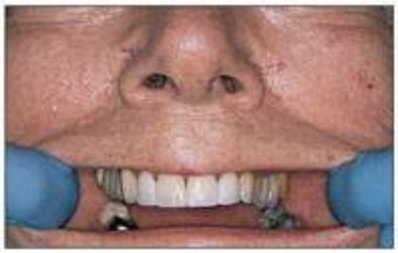


Fig 17-1 Pulling the upper lip parallel with the interpupillary line to examine the incisal plane. Note the sigmoid curve of the incisal plane.

Diagnostic aids

Mock-ups

With an understanding of esthetic diagnostic parameters, the dentist can use several diagnostic methods to develop a treatment plan that will predictably result in success. Preparation and waxing on a diagnostic cast are sometimes helpful to the dentist, especially when the veneers are intended to lengthen teeth, close spaces, or correct malaligned teeth. However, a diagnostic wax-up is not especially helpful in giving the patient a preview of the expected esthetic outcome. This can be accomplished more effectively by one of the intraoral techniques.

A mock-up of the desired result can be accomplished with direct placement of composite in the patient's mouth to simulate the desired outcome (Fig 17-2). Because the teeth are not acid etched before the intraoral mock-up, the composite can easily be removed from the tooth. This technique is especially helpful for simple procedures, such as diastema closure.



Fig 17-2*(a)* Diastema. *(b)* Diagnostic intraoral mock-up of diastema closure.

However, when major changes are being considered, such as lengthening several teeth, direct mock-up is too time-consuming (Fig 17-3a). In this situation, a diagnostic wax-up is done on a preoperative cast, and a matrix is made. Bis-acrylic provisional material is placed in the matrix, which is placed over the teeth intraorally and allowed to cure (Fig 17-3b). Once the excess flash is cleaned from the teeth, the patient may preview the projected outcome (Fig 17-3c). At this point, changes can be made (eg, shortening or lengthening of the teeth) until the patient is comfortable and satisfied. An impression can then be made of the corrected mock-up in the mouth, and the subsequent cast will serve as a reference cast for the laboratory technician. This same technique will be recommended as a preparation guide to insure the proper depth of reduction during treatment (Fig 17-3d). Another method is to construct a resin composite shell or overlay on a diagnostic cast. The shell is placed intraorally to allow the patient to visualize the projected result. This is an especially helpful diagnostic tool for the patient with excessive gingival display. The composite shell can extend over the gingiva to demonstrate the esthetic effect of surgically lengthening the clinical crowns (Fig 17-4).



Fig 17-3*(a)* A patient presents with pathologic wear and excessive anterior spacing requiring modification beyond a direct mock-up. *(b)* Bis-acryl provisional material is placed in a matrix made from a diagnostic wax-up and allowed to cure on unprepared teeth. *(c)* The patient can evaluate the esthetic alterations. *(d)* The matrix can also serve as a reduction guide for veneers.

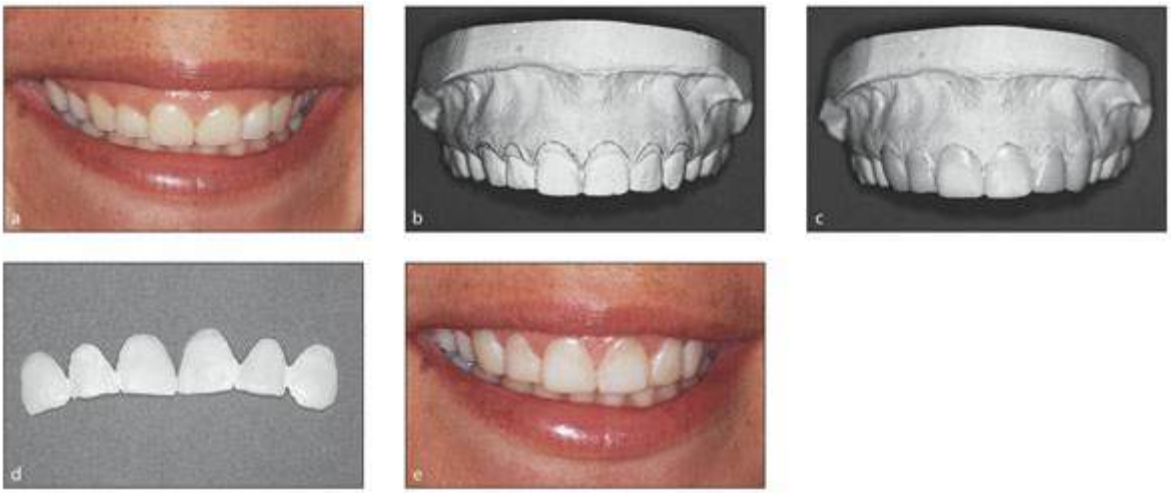


Fig 17-4*(a)* Short clinical crowns resulting from excess gingival coverage. *(b)* Preoperative study cast with lines drawn to aid in the fabrication of a diagnostic resin composite overlay. *(c and d)* Diagnostic resin composite overlay. *(e)* Diagnostic resin composite overlay demonstrating the benefit of crown lengthening of the anterior teeth.

Commonly, a preview of the projected outcome is achieved with the provisional restorations (Fig 17-5). A diagnostic wax-up is accomplished, and a stent is fabricated. After the preparations are completed and the final impression is made, the provisional restorations are constructed using the stent. Once the patient is satisfied with the provisional restorations, a cast of the arch, with the provisional restorations in place, is made and sent to the laboratory to serve as a blueprint for the definitive restorations.



Fig 17-5*(a)* Patient's preoperative presentation demonstrating multiple diastemas and reverse smile line. *(b)* Diagnostic wax-up. *(c)* Silicone matrix of wax-up. *(d)* Enamel-depth bonded veneer preparations with composite restorations removed. *(e)* Bis-acryl provisional restorations. *(f)* IPS e.max (Ivoclar Vivadent) definitive veneer restorations (Porcelain veneers by Lebeau Dental Laboratory, Renton, Washington.)

Computer imaging

Another diagnostic method involves computer imaging of the patient's smile, making the desired esthetic changes on the screen. This provides both the patient and the dentist with a preview of the expected result (Fig 17-6). However, imaging is limited because it does not allow for a dynamic evaluation in the same way that an intraoral mock-up allows the patient to experience the projected changes. Because many dentists do not have the in-office capability for computer imaging, the patient or a photograph of the patient may be sent to a location that offers the service.

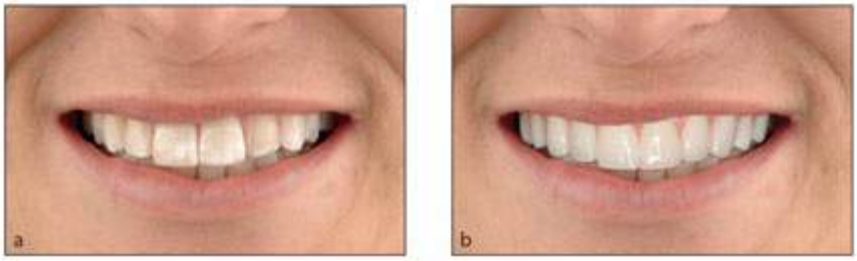


Fig 17-6*(a)* Preoperative presentation of a patient considering veneers to improve color and contour of her smile. *(b)* Computer simulations of smile changes (Envision A Smile) can help the patient visualize the final results.

Photographs

Preoperative photographs offer another important diagnostic aid; they document the preoperative condition and aid the technician in the fabrication of the veneers. The series should include a full-face smile; a retracted frontal image of maxillary and mandibular teeth in occlusion; a retracted frontal view with a shade tab held directly beneath the incisal edges of the maxillary incisors; a retracted close-up view of the teeth to be veneered, with and without a shade tab; and a postpreparation view of the teeth to be veneered, with a shade tab. Specialized dentin shade tabs are available to mimic dentin coloration (Fig 17-7). In addition, other photographs, such as a profile or a view of the intraoral diagnostic mock-up, should be included if they can benefit the laboratory technician.



Fig 17-7 Shade tab held under the incisal edges of the prepared teeth. This shade guide is specially designed to demonstrate the dentin color of the preparation.

Single Veneer

Perfectly matching a full-coverage restoration to an adjacent natural central incisor is an extremely difficult restorative procedure. Commonly, the porcelain veneer is the restoration of choice in this situation. If the tooth to be restored is not significantly discolored, the porcelain veneer is an excellent restorative option. The major advantage of the single porcelain veneer restoration is the dentist's ability to increase or decrease the value of the restoration with the bonding resin cement (Fig 17-8).

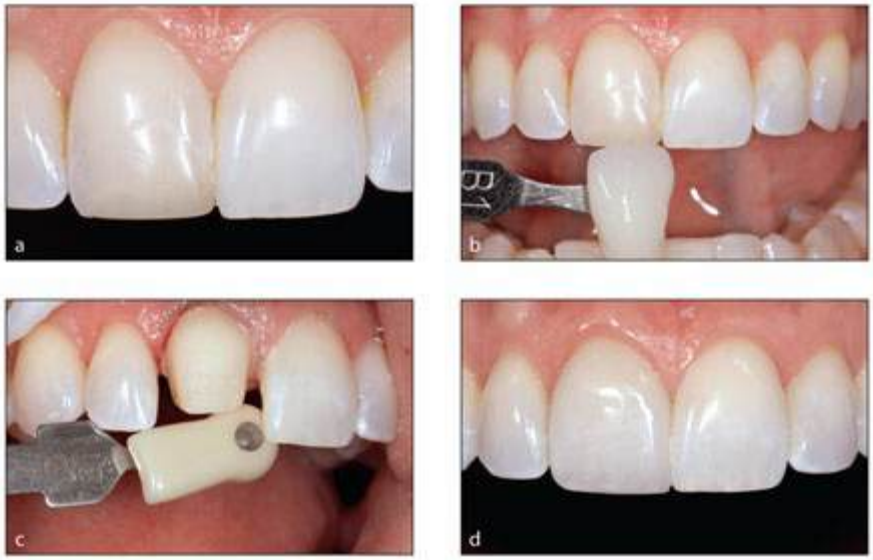


Fig 17-8*(a)* Right central incisor fractured and repaired with composite that has become stained. *(b)* Multiple shade photographs are made before tooth preparation to capture the texture, chroma, and maverick colors of the teeth. *(c)* Preparation color captured with dentin shade guide. *(d)* Definitive porcelain veneer restoration of the right central incisor.

Multiple Veneers

When the clinician has the option of veneering multiple anterior teeth, the problem of shade matching is minimized. It is easier to deal in even numbers when veneers are placed on anterior teeth. It is much simpler to veneer two central incisors than to attempt to match a veneer to a natural tooth. Therefore, the chances of obtaining an optimally esthetic result are enhanced when two, four, six, or eight veneers are placed.

An option that is commonly chosen is the placement of six veneers from canine to canine. In Fig 17-9, the anterior teeth are brighter and bolder, while the buccal corridor appears to become darker. This accentuates the anterior teeth and commonly creates the unpleasant illusion that the anterior teeth are larger and longer

as well as brighter. This does not usually occur when only the four incisors are veneered (Fig 17-10). Therefore, when veneers are planned for the incisors but there is no esthetic or functional requirement for canine veneers, the esthetic result is enhanced when the canines are left unrestored. However, when all six anterior teeth require veneers, consideration should be given to veneering one or more posterior teeth on each side, depending on the posterior extent of the smile and the maxillary arch form.



Fig 17-9 Porcelain veneers bonded on all maxillary anterior teeth. Note the apparent separation of anterior and posterior segments of the mouth because of the boldness of the porcelain veneers.

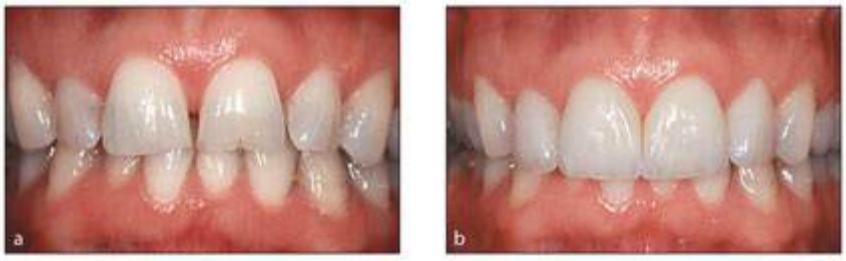


Fig 17-10*(a)* Preoperative view of maxillary central and lateral incisors. *(b)* Porcelain veneers bonded on the maxillary central and lateral incisors to blend esthetically with the maxillary canines. (Porcelain veneers created by Steve McGowan, CDT, Kenmore, Washington.)

Failure

The most common cause of failure of porcelain veneers is fracture.[5,10–12] Clinical studies report a modest 0% to 5% failure rate due to fracture. Higher fracture rates (7% to 14%)[12] were noted in cases with unfavorable occlusion, significant parafunction, large dentin bonding surfaces, and bonding to existing restorations. In a 15-year review, Friedman[11] classified the fractures into three categories: *(1)* static, *(2)* cohesive, and *(3)* adhesive. When a segment of a veneer fractures but remains intact, it is defined as a *static fracture* (Fig 17-11). These failures are caused by excessive loading or polymerization shrinkage. The key factors are the internal fit of the ceramic restoration and the amount of unsupported porcelain. The crack propensity is inversely proportional to the internal fit of the veneer. An internal fit

discrepancy of 100 μm or less will minimize internal stress and prevent static fracture.[13,14] *Cohesive fractures* occur within the body of porcelain due to tensile loads from excessive functional or parafunctional loading (Fig 17-12). A cohesive failure results in the loss of a fragment of ceramic, requiring repair or replacement of the veneer. Enamel imparts stiffness to the tooth much like a metal coping does for a metal-ceramic crown. Removal of the enamel negatively affects the stress-strain distribution of the subsequent veneer. This leads to an increase in flexure under load and, ultimately, cohesive fracture.[14] The most important areas in which to maintain enamel are the incisal and cervical areas. Lack of adhesion in those areas produces higher stress on loading and increased risk of cohesive fracture.[15] Finally, an *adhesive fracture* is due to a failure of the bonding interface between the porcelain/luting composite and the tooth structure (Fig 17-13). It is a result of a weak bond or severe occlusal loading. Friedman[11] reports that 86% of the adhesive fractures occurred at a resin-dentin interface.

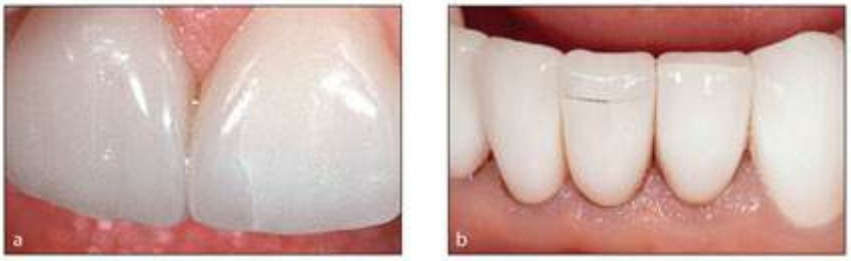


Fig 17-11 *(a)* Static fracture of a porcelain veneer. *(b)* Horizontal static fractures of mandibular incisors. These fractures are obvious to the patient because of the stain and refraction of light. A vertical static fracture of the canine is not as evident.



Fig 17-12 Cohesive fracture of maxillary left central incisor veneer due to excessive force.

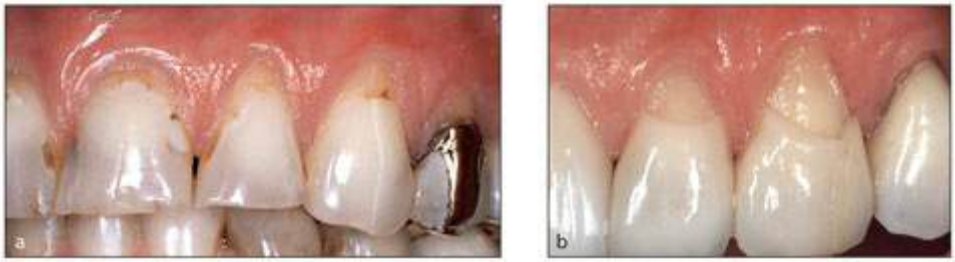


Fig 17-13 *(a)* Exposed dentin surfaces provide a poor bonding substrate for veneers, increasing

the risk of fracture. *(b)* Adhesive failure of dentin-bonded veneer.

On rare occasions, a porcelain veneer will debond. When this happens, it is important to determine at which bonded interface the failure occurred. If the luting composite remains on the tooth, the failure is likely due to either inadequate etching of the veneer or the use of silane past its expiration date (Figs 17-14a and 17-14b). The stated shelf life of silane is approximately 1 year when it is refrigerated, but it is known that silane efficacy decreases with time. If the luting composite remains on the inside of the veneer, then there was a problem with either the bonding materials, the placement technique, or the bonding substrate (Figs 17-14c and 17-14d). Veneers that are bonded to a predominantly dentinal substrate have a significantly greater likelihood of debonding than veneers that are predominantly bonded to enamel.[10]



Fig 17-14*(a and b)* Debonding of the veneer leaving resin composite bonded to the tooth is a result of inadequate porcelain etching or contaminated silane. *(c and d)* Resin composite attached to the debonded veneer indicates a predominantly dentinal substrate.

Finally, marginal staining and leakage are common causes of failure of porcelain veneers.[1–5] The staining is caused by an influx of oral fluids containing chromogenic bacteria or organic stains. The composite-tooth interface is the primary site for the leakage. It has been noted that these areas of microleakage are always devoid of enamel.[11]

Tooth Preparation

The preparation of teeth for porcelain veneers is usually uncomplicated when the basic principles are understood and followed. Historically, veneers were placed on unprepared teeth, making the technique conservative and reversible. This led to overcontoured restorations, gingival irritation, and high failure rates[16,17] (Fig 17-15). The stress concentration is less severe on veneers fitted to prepared teeth.[18] In addition, the preparation removes the aprismatic and hypermineralized enamel

layers, which can be more resistant to acid etching.[19] Today, practitioners should focus on maintaining the preparation completely in enamel to maximize the resin bond strength and decrease the tensile stresses in the porcelain.[15,20] Even with the newest-generation bonding agents, the bond strength of porcelain with resin cement to enamel is far superior to the bond strength of porcelain with resin cement to dentin. Fractures, microleakage, and debonding are all failures that can be linked to preparations situated in dentin.[11]



Fig 17-15*(a)* Lack of preparation may create overcontoured restorations. *(b)* Ultraconservative full-veneer preparations resolve contour issues yet still maintain an enamel bond.

Anterior teeth

Gingival finish lines

When preparing teeth for porcelain veneers, it is essential to remember that the strongest and most predictable bond is to enamel. The more dentin that is exposed during the preparation, the poorer the bond of the veneer and the poorer the ultimate stress distribution during function.[19] For maxillary porcelain veneers, the gingival margin of the veneer should routinely be placed at the gingival crest or slightly subgingivally. A primary goal of the preparation is to have all margins on sound enamel, because the stress distribution in the veneer is much improved when all margins are bonded to enamel. The amount of exposed dentin in the central portion of the veneer preparation becomes much less important when all of the margins are bonded to enamel.[19] Because the enamel is only approximately 0.3 mm thick at 0.5 mm coronal or incisal to the cementoenamel junction,[21,22] it is difficult to obtain adequate preparation depth while preserving the enamel. There is also a significant decrease in the average enamel thickness as the patient ages. This decrease in enamel for patients aged 50 years and older increases the risk of dentin exposure and a less reliable bond.[23]

In a routine preparation, the facial enamel reduction must be exquisitely controlled. This cannot be accomplished freehand. It may only be accomplished with

the use of a depth guide.[24–27] The authors use single-head depth-cutting burs, although a dimpling technique with a 1-mm round bur has proven effective at preserving enamel[25,26] (Fig 17-16). After a 0.3-mm gingival depth cut and a 0.5-mm midfacial depth cut, the enamel is uniformly removed with a round-ended, cylinder diamond. Unless the teeth to be veneered are dark (low value), the gingival preparation should not routinely extend more than minimally into the gingival sulcus. In the midfacial area, the enamel thickness ranges from 0.8 to 0.9 mm on average. The preparation should deepen from the midfacial and taper into the incisal, increasing to 0.7 mm (Fig 17-17). This allows the technician to effectively disguise the transition from supported to unsupported porcelain.



Fig 17-16 Depth-cutting burs of different designs and depths.

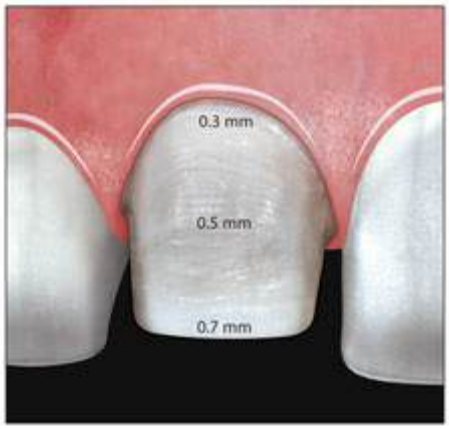


Fig 17-17 Average facial reduction for enamel-bonded veneer.

When mandibular anterior teeth are prepared for porcelain veneers, the considerations are different from those for maxillary teeth. In most patients, the gingival half of the mandibular incisors remains covered by the lower lip at all times, resulting in no esthetic display. In addition, the marginal gingiva of the mandibular anterior teeth is commonly thin, and the gingival sulcus is narrow and shallow, making the placement of gingival retraction cord very difficult. For these reasons, the gingival margins of the preparations for mandibular anterior teeth are ideally placed at least 1.0 mm incisal to the marginal gingiva.

The long-term predictability of a porcelain veneer is directly proportional to the amount of enamel substrate in the preparation.[28] Preparation guides have been proposed as a technique to ensure that the tooth is properly prepared. Malpositioned and/or worn teeth, frequently the teeth requiring veneers, make use of a guide for preparation depth difficult (Fig 17-18a). The depth of reduction is formulated from the final contour and length of the tooth as it will be restored and not the preoperative position (Figs 17-18b and 17-18c). Traditional preparation guides only ensure that enough tooth structure has been removed for the laboratory. They do not limit the reduction, which is the key in veneer preparation. Aesthetic preevaluative temporaries (APTs)[29] assure enough reduction for the laboratory but also allow the dentist to limit the damage caused by overreduction. A silicone matrix is fabricated from a diagnostic wax-up or an approved intraoral mock-up (Fig 17-19a). After the patient is anesthetized and the teeth dried, bis-acrylic resin is injected into the matrix. It is seated and allowed to fully cure (Fig 17-19b). An important step in creating ideal positioning and thickness of the mock-up is allowing the acrylic to escape from the matrix when it is seated. Because the teeth are not prepared, the simulation will in most cases be very thin. If the acrylic is not allowed to flow easily from the matrix, it will be trapped, creating additional facial or incisal bulk. Reduction cuts into this oversized matrix will not provide accurate reduction. Slits in the matrix or vent holes from the facial will assist in encouraging the acrylic to flow from the form. The mock-up is not removed from the teeth but instead allowed to lock into the embrasures. This allows the dentist to perform ideal reduction of the mock-up before removing it from the teeth (Figs 17-19c and 17-19d). Once removed, the proper depth cuts will be found on the teeth, thus assuring ideal reduction without the inherent risk of overreduction created with traditional reduction guides (Figs 17-19e to 17-19h).



Fig 17-18*(a)* Malaligned teeth make preparation guides essential. *(b)* Reduction guide fabricated from the completed wax-up. *(c)* Reduction guide from the final waxed contours provides little assistance in determining proper reduction. The guide becomes valuable after tooth reduction, which may be too late to preserve enamel.



Fig 17-19*(a)* Clear polyvinyl siloxane matrix produced from a modified wax-up. (The camera flash imparts a blue tinge to the matrix.) A clear matrix serves as a reduction guide as well as the matrix for composite provisional veneer fabrication. *(b)* Bis-acryl resin is allowed to cure, demonstrating ideal final contours and locations of tooth structure in buccoversion. *(c)* Preparation depth cuts into the matrix provide an ideal guide for ultraconservative veneer preparation and lessen the likelihood of overpreparation. *(d)* The guide is removed and depth cuts visualized. *(e)* Completed preparations. *(f)* Traditional reduction guide demonstrates uniform preparation depths. *(g)* Definitive porcelain veneer restorations. *(h)* Smile at 4-year follow-up.

Interproximal contact area

For the purposes of veneer fabrication and placement, it is important that the preparation not be finished in the interproximal contact area. When the margin is stopped in the interproximal contact area, it is difficult to accurately capture the margin in the final impression. Additionally, veneer fabrication is more difficult for the laboratory technician, and bonding and finishing procedures are more difficult for the dentist. For these reasons, the preparation must either stop facial to the interproximal contact area (Fig 17-20a) or extend completely through the contact area to the lingual surface[22,30] (Fig 17-20b).

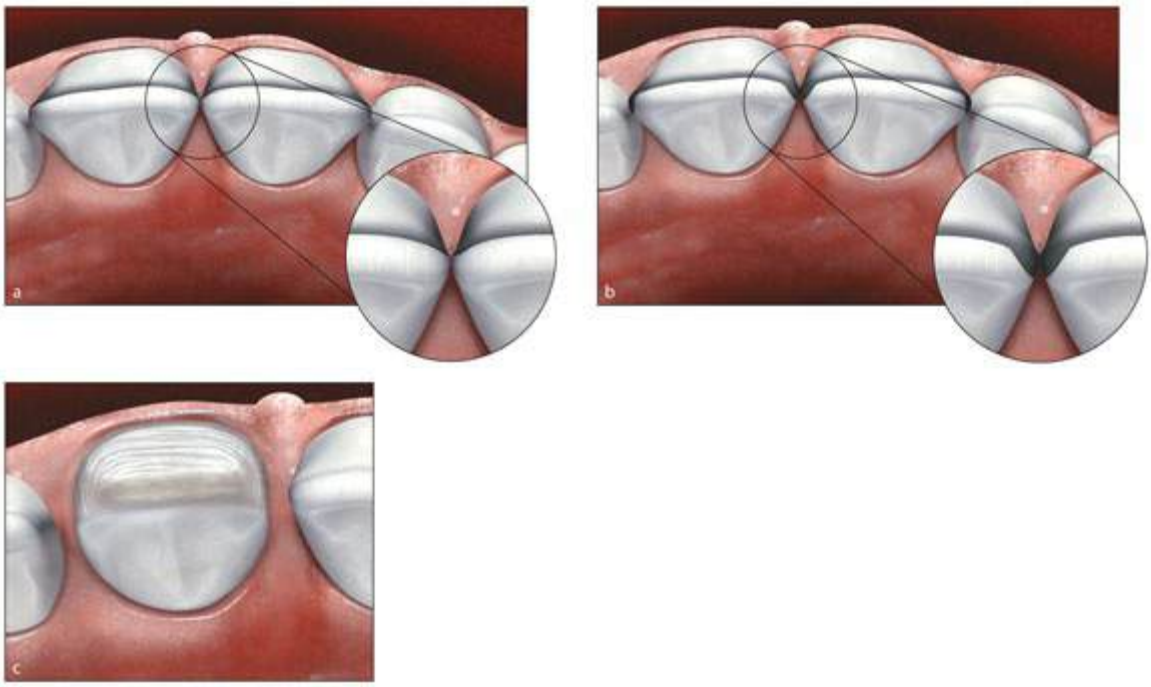


Fig 17-20 *(a)* Preparation extending slightly facial to the interproximal contact area. *(b)* Preparation extending completely through the interproximal contact area to the palatal surface. *(c)* Completed veneer preparation through the interproximal contacts. This preparation is commonly referred to as a *full veneer*.

A major long-term problem with porcelain veneers is marginal staining.[5,11] This staining is most apparent at the proximal margins. Therefore, the current trend is to extend the preparation through the interproximal contacts (Fig 17-20c) from the mesial aspect of the canine to the mesial aspect of the contralateral canine. There is no esthetic need to extend the preparation through the contact areas on the distal surfaces of the canines. By extending the preparation through interproximal contacts in areas other than the distal contacts of canines, any proximal marginal staining is concealed. With dark teeth (eg, tetracycline-stained teeth), it is imperative that the preparation extends completely through the interproximal contact area to the lingual surface (Fig 17-21). This decreases the risk of dark shadows appearing around the periphery of the veneer.[30]



Fig 17-21 *(a)* Traditional interproximal extension during veneer preparation will exhibit the tooth discoloration. *(b)* Placing the finish line to the lingual eliminates the chance of interproximal discoloration. *(c)* Preparation-shade photographs allow the technician to modify the ceramic to mask the severe tetracycline stains. *(d)* Completed veneers hide the significant discoloration while remaining lifelike.

Incisal edge

There is ongoing debate regarding the need to cover the incisal edge of a maxillary tooth with the porcelain veneer. If there is no esthetic requirement to change the incisal shape or length, and there is adequate remaining incisal tooth structure after facial reduction, the incisal margin may be terminated at the facioincisal line angle (Fig 17-22a). This preparation is most commonly indicated for linguoverted teeth. An alternative is the "window preparation" (Fig 17-22b). This preparation provides the most protection for the veneer during function; however, it results in a visible composite-veneer interface at the facioincisal margin. This preparation is most commonly indicated for the maxillary canine that has the correct length and incisal edge configuration and has a large lingual wear facet extending to the incisal edge. Because it is preferable not to place a veneer margin in a wear facet, the window preparation is appropriate. Neither of these preparations can be used, however, when the tooth is being lengthened incisally with the porcelain veneer. When the maxillary incisal edge is not reduced during the preparation, the veneer is inevitably thicker, and the incisal edge is too round buccolingually, lacking a more natural, sharp facioincisal line angle. Additionally, veneers that do not cover the incisal edge are significantly more difficult to orient correctly during bonding.

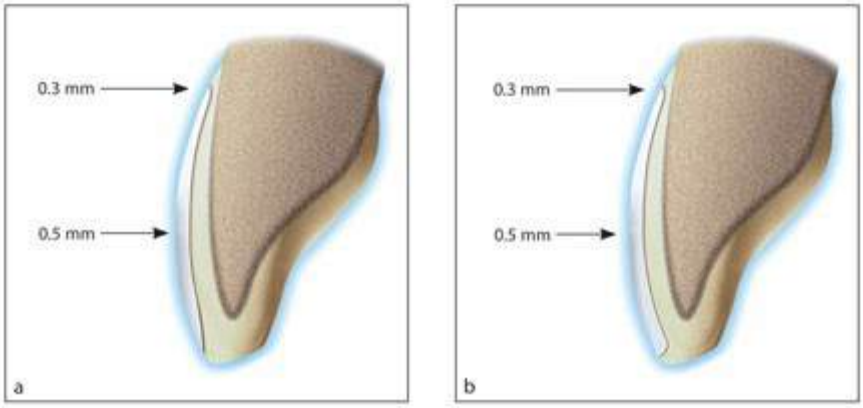


Fig 17-22*(a)* Veneer preparation that does not cover the incisal edge of the maxillary anterior tooth. *(b)* Window preparation on a maxillary anterior tooth.

The most universally applicable preparation, which allows the technician to incorporate the natural incisal translucency into the veneer, requires incisal edge reduction. If the incisal edge of the maxillary tooth is to be covered with the veneer, there must be room for at least 1 mm of porcelain over the incisal edge. However, a 1.5- to 2.0-mm incisal edge reduction from the final incisal edge position (Fig 17-23) allows the laboratory technician to provide maximum esthetics in the incisal third of the veneer. Therefore, a preoperative determination of the ideal incisal edge position must be made before the tooth is prepared. A diagnostic wax-up and preparation guide[29] can be helpful in this evaluation. In the case of a worn or fractured tooth, more incisal porcelain length is acceptable if the veneer remains bonded to enamel and the fit will provide a cement thickness of 100 µm or less.[13,31]

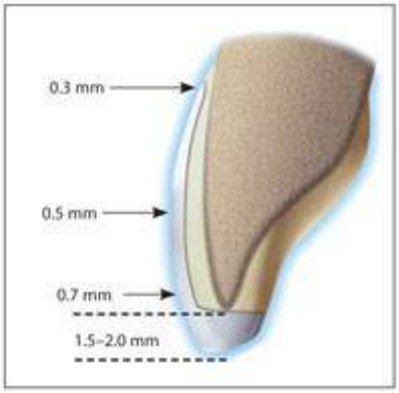


Fig 17-23 Standard veneer preparation for maxillary anterior teeth. Incisal reduction is 1.5 to 2.0 mm.

Traditionally, the lingual finish line configuration was a chamfer. This placed the thin veneer in the high–tensile stress lingual cingulum area of the tooth, significantly weakening the restoration and creating a higher risk of crack formation on the

lingual.[32] Today, the incisal edge should be flattened, leaving a butt finish line configuration on the lingual surface.[33] It is important to round the sharp facioincisal line angle of the preparation to prevent stress concentration in the bonded veneer and to allow for an undetectable transition from unsupported to supported porcelain (Fig 17-24). One notable exception to this rule is made for severely worn teeth. If the amount of enamel incorporated into the final preparation is predictive of the long-term success of the veneer, severely worn teeth would have a poorer survival rate. Creating a long chamfer on the lingual of worn teeth may access additional enamel. An in vitro report concluded that, on severely worn teeth, a palatal chamfer margin significantly increased the final veneer load-to-failure values compared with a shoulder finish line[34,35] (Fig 17-25).



Fig 17-24 Inadequate depth of preparation in the incisal third leads to an unsightly show-through of the preparation, which is seen to a greater degree on the left central incisor.



Fig 17-25*(a)* Significant wear on maxillary anterior teeth. Orthodontic intrusion aligns gingival architecture and creates restorative space. (Orthodontics by Jay Gibson, San Antonio, Texas.) *(b)* Parafunctional wear is evident on the lingual surfaces. *(c)* Full-veneer preparations. *(d)* Lingual finish lines carried below wear facets to access more bondable enamel. *(e)* Completed veneers. *(f)* Additional bonding to enamel is gained with a lingual chamfer extension, improving the strength of the restoration.

If the practitioner chooses to maintain the interproximal contact, the proximal portion of the preparation should follow the papilla and extend slightly under the interproximal contact to ensure coverage of the tooth in this area (Fig 17-26). This extension is called an *elbow preparation*. If the tooth is not prepared in this manner or extended through the contact, the darker, unprepared triangle of natural tooth structure results in less-than-optimal esthetics.

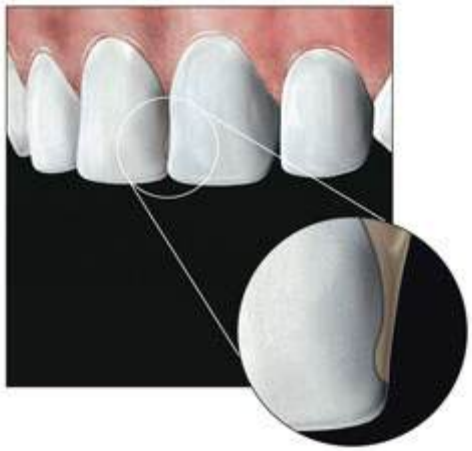


Fig 17-26 Extension of the veneer preparation into the subcontact area for improved esthetics.

The incisal edges of mandibular anterior teeth should be routinely covered with porcelain at least 1.5 to 2.0 mm thick (Fig 17-27). Incisal edge flattening with a resultant lingual butt margin is also recommended for mandibular anterior teeth. Again, the facioincisal line angle of the preparation must be rounded.

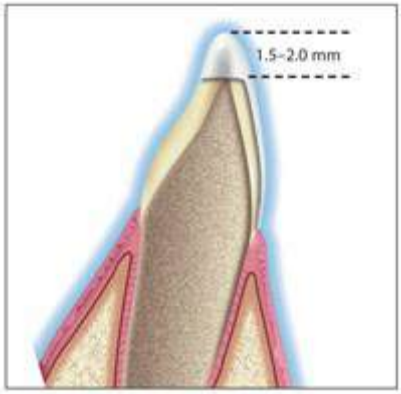


Fig 17-27 Veneer preparation on a mandibular incisor. There should be at least 1.5 to 2.0 mm of incisal porcelain in the definitive restoration, and the gingival margin is at least 1.0 mm incisal to the gingival crest.

When multiple teeth are being prepared, incisal reduction should be symmetric (eg, both prepared lateral incisors should be the same length) for more uniform

esthetics in the definitive restorations. After completion of all veneer preparations, the dentist should stand in front of the patient, retract the patient's lips, and confirm that the preparations are symmetric and the incisal edges parallel to the horizon.

Overlapping teeth

When overlapping teeth are being prepared, the paths of insertion of the veneers must be considered. Where there is significant misalignment of the teeth, it is usually impossible to develop paths of insertion for the veneers without extending the preparations completely through the interproximal contact areas. In addition, significant alteration of the interproximal contact position and straightening of the facial surfaces is usually facilitated when the teeth are prepared through the contact (Fig 17-28).

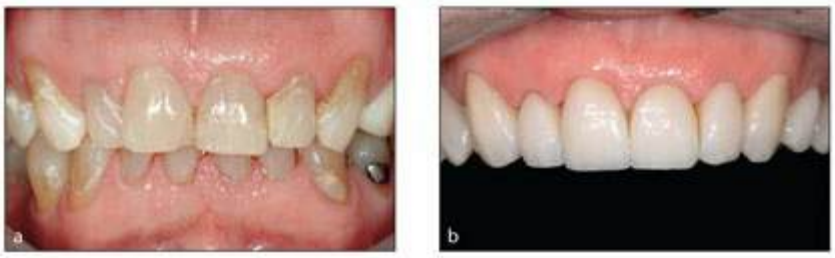


Fig 17-28*(a)* Correction of mild to moderate overlapping requires preparation through the contacts to allow contour alteration. *(b)* Porcelain veneer restorations demonstrating ideal alignment. Gingival contours are irregular because orthodontic alignment was not completed.

Space closure

Preparation of teeth for space closure presents a unique situation. To obtain smooth lingual contours, the proximal finish lines adjacent to the space must be made more lingual. The wider the space to be closed, the more lingually the tooth must be prepared (Fig 17-29). Also, the proximal margins adjacent to the space must be positioned subgingivally so that the gingival contours of the veneers minimize the black triangle between the bottom of the interproximal contact area and the tip of the papilla. However, the gingival margin of the veneer should be no closer than 2.5 mm to the crest of the alveolar bone.[36] As teeth separate, the scallop of the interproximal bone lessens, resulting in a wide, flat papilla. Extending the preparation subgingivally and lingually enables the technician to alter the interproximal contour of the veneers and reshape the papilla (Fig 17-30).

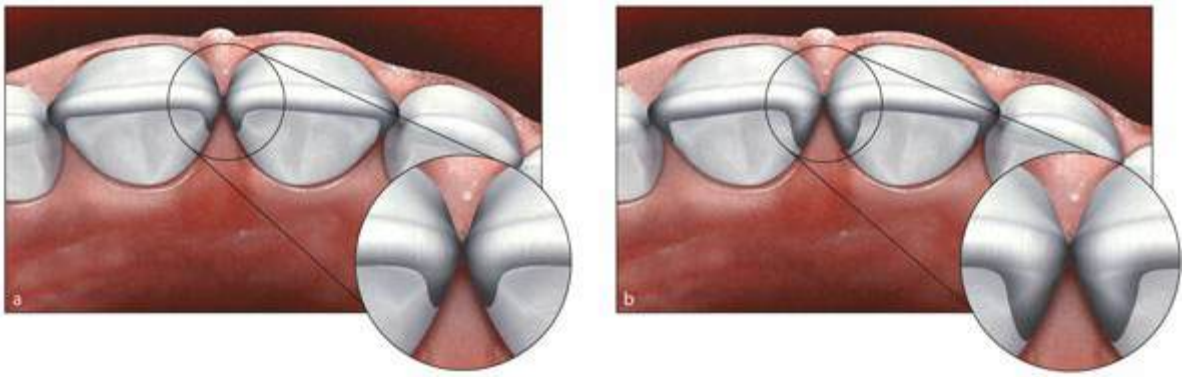


Fig 17-29*(a)* Porcelain veneer preparations for closing a small to moderate diastema. *(b)* Porcelain veneer preparations for closing a large diastema. The preparations involve more of the lingual surfaces of the central incisors to develop adequate lingual contours.



Fig 17-30 Overcontouring the veneers allows alteration of the interproximal papilla contour.

Premolars

The considerations for preparing maxillary premolars are similar to those for anterior teeth. The veneer preparation of the maxillary premolar may end on the facial surface of the tooth if the functional and esthetic requirements are met (Fig 17-31a). Ideally, the occlusal margin should be placed so that it is not in an occlusal contact area and so that the opposing cusp does not function across it. If the anterior guidance immediately disoccludes the posterior teeth in mandibular excursive movements, the occlusal margin can be placed anywhere on the lingual incline of the buccal cusp, as long as there is no occlusal stop directly on the margin (Fig 17-31b). However, when group-function occlusion is present, it may be necessary to extend the occlusal margin to the central groove. In this circumstance, occlusal reduction must be adequate to allow for an onlay of at least 2-mm-thick porcelain over the buccal cusp, blending to the 0.3- to 0.5-mm reduction of the facial surface (Fig 17-31c). In mandibular premolars, because the esthetic cusp is also the functional cusp, the tooth must generally be prepared for a porcelain onlay rather than simply for a buccal porcelain veneer. The interproximal contact is rarely broken in second premolars. Esthetics may dictate extending through the mesial contact of first

premolars (Fig 17-32).

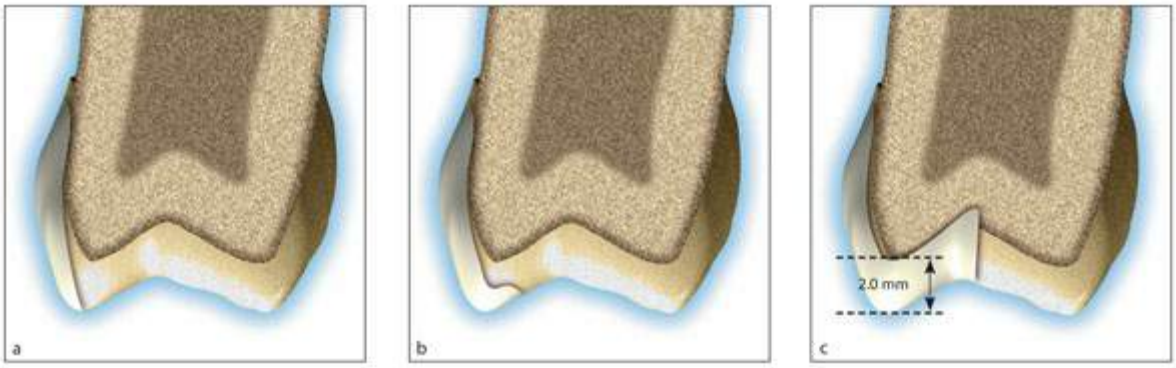


Fig 17-31 *(a)* Porcelain veneer preparation that does not cover the buccal cusp of the maxillary premolar. *(b)* Veneer preparation overlapping the buccal cusp. There should be no occlusal stop directly on the veneer margin. *(c)* Porcelain veneer preparation that covers the buccal cusp of the maxillary premolar. Note the 2.0 mm of occlusal reduction on the buccal cusp.

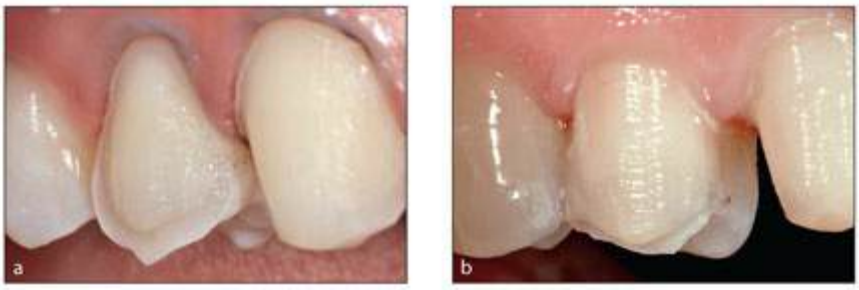


Fig 17-32 *(a)* Typically, the interproximal contact remains facial to the contact for maxillary premolars. *(b)* When esthetics or function dictates, the mesial preparation of a maxillary first premolar will be extended to the lingual.

Existing restorations

Existing resin composite restorations can complicate the veneer preparation. Ideally, no margins should be finished on an existing resin composite restoration. One clinical study reported a significantly higher failure rate for porcelain veneers placed over existing resin composite restorations.[17] Bonding effectively to an existing restoration is questionable, particularly if it has been in place for any period of time. Water sorption, unsilanated surfaces of filler particles, and limited unpolymerized resin in the set composite lead to a significant decrease in bond strength.[10,24] The resin composite restoration should be removed during tooth preparation for the veneer, and the missing tooth structure should be replaced as part of the porcelain veneer[37] (Fig 17-33). While proper surface conditioning may eventually provide higher fracture strengths that eliminate the necessity of removing the aged composite restorations, the difficulty of maintaining the existing composite

restorations associated with the veneers makes it undesirable.[38] Furthermore, attempting to remove a failing Class 3 composite restoration at the margin of a veneer is a risky procedure that restorative dentists should avoid. However, small areas of resin composite can remain centrally in the preparation to serve as an undercut block-out material.



Fig 17-33*(a)* Teeth with Class 3 composite restorations are initially prepared as if the restoration is nonexistent. *(b)* Class 3 restoration demonstrated after initial tooth reduction. *(c)* The restoration is removed. *(d)* Preparation is modified to allow the veneer to incorporate the missing tooth structure. *(e)* Veneer preparations on the central and lateral incisors replacing composite restorations.

Fractured incisal edge

A unique indication for the porcelain veneer is the restoration of the fractured incisal edge of an incisor (Fig 17-34). Although there is little clinical data to support the use of veneers for this purpose, laboratory data are favorable.[31,37] The limits for the amount of missing tooth structure that can be replaced with a porcelain veneer are not known. However, anecdotal evidence seems to indicate that approximately 50% of the clinical crown can be replaced with a porcelain veneer when the preparation on the remaining tooth structure is in enamel. Because the veneered tooth does not require endodontic treatment for restorative reasons or the placement of a post and core, it is a highly desirable alternative to previous methods of restoring the fractured tooth. The preparation for this type of veneer extends to near the gingival crest on both the facial and lingual surfaces.



Fig 17-34 *(a)* Fractured mandibular right central incisor. *(b)* Preparation of the fractured mandibular right central incisor for a porcelain veneer. *(c)* Porcelain veneer bonded on the mandibular right central incisor. (Porcelain veneer created by Gilbert Young, CDT, Plano, Texas.)

Noncarious cervical lesions

Noncarious cervical lesions (NCCLs), caused by erosion, abrasion, and/or abfraction, provide a different, more sclerotic dentin substrate for bonding than surfaces not affected by these processes. Advancements in adhesive systems have improved adhesion to sclerotic dentin, with some studies showing no difference in adhesion to sclerotic dentin compared with unaffected dentin, while other studies demonstrate reduced bonding efficacy to sclerotic dentin compared with unaffected dentin. A number of factors can influence the quality of bonding to the sclerotic dentin, including the specific adhesive system, treatment of the dentin surface, and the placement technique.[39–46] Adhesion to NCCLs provides an additional challenge to providing bonded restorations. Therefore, the best treatment solution is to surgically cover these areas with connective tissue grafting.[47] (See chapter 15 for more on tissue grafting of NCCLs.) This results in more natural gingival architecture and allows the gingival margin of the veneer preparation to end on enamel. However, if the surgical option is not chosen and the esthetic demands require that the root be covered with a veneer, the preparation must extend onto the root surface. The patient should be advised that the risks of staining, microleakage, and/or veneer fracture in the gingival third are much greater because of the less-than-ideal bonding.[15] Full-coverage restorations must also be considered.

Impressions

When an impression is made of the maxillary teeth, retraction cord is placed to expose all gingival margins. This step is generally not necessary with mandibular teeth because the preparations are at least 1.0 mm incisal to the marginal gingiva. An accurate impression material, such as polyvinyl siloxane, polyether, or reversible hydrocolloid, is then used to make the final impression. See chapter 18 for more

information on taking impressions for esthetic indirect anterior restorations.

Provisional Restorations

The placement of provisional restorations over veneer preparations is an integral step in the predictable placement of porcelain veneers. Provisional restorations not only improve interim esthetics and decrease sensitivity, but they also provide essential diagnostic information, including veneer color, shape, length, and incisal edge configuration, that cannot be obtained in any other way.[48] When provisional restorations are being placed on one or two teeth, the procedure is best accomplished with freehand placement of composite. A small area in the incisal third of each tooth is etched with phosphoric acid for 15 seconds, washed, and dried. Adhesive resin (not self-etching adhesive) is placed over the entire preparation and light cured. A large increment of resin composite is then patted into place with correct contours, the gingival margins are smoothed with an explorer tip or other fine-tipped instrument, and the provisional restoration is light cured. There should be no overhanging resin composite at the margins, and the provisional restoration should require virtually no adjustment.

When provisional restorations are being placed on multiple teeth, it is preferable to use a clear matrix made on a preoperative diagnostic cast. A diagnostic wax-up is commonly required to change tooth length, alignment, and/or incisal edge configuration (Figs 17-35a and 17-35b). If the clear matrix is made of a plastic stent material, the diagnostic wax-up cast must be first duplicated (Fig 17-35c). If the matrix is made of a clear polyvinyl siloxane bite registration material, it can be made directly on the diagnostic wax-up cast. The teeth are spot etched with 30% to 40% phosphoric acid in the incisal third (Fig 17-35d), washed, and air dried. The entire preparation is covered with adhesive resin, which is then light cured (Fig 17-35e). A self-etching primer system should not be used because the provisionals will be bonded to the entire preparation and are difficult to remove without altering the preparations. The facial and incisal areas in the clear matrix are filled with resin composite (Fig 17-35f), and the matrix is placed over the prepared teeth (Fig 17-35g). The gingival two-thirds of the matrix is shielded from the polymerization light, and the incisal one-third is polymerized with the light for 10 seconds per tooth (Fig 17-35h). The gingival two-thirds is then lightly cured for 0.5 to 1 second per tooth (Fig 17-35i). The matrix is gently teased away from the tooth at the gingival margin to ensure that the resin composite does not stick to the matrix (Fig 17-35j). If the resin composite sticks to the matrix, the matrix is returned to place and the gingival

two-thirds is polymerized again for 0.5 second per tooth. The matrix is removed, and the excess partially cured resin composite is first removed proximally and lingually with a no. 12 or 12B scalpel blade (Fig 17-35k). Floss and a floss threader are then used in each gingival embrasure to ensure patency and that there are no overhangs (Fig 17-35l). The gingival margins are then carved with the no. 12 or 12B scalpel blade. If small areas of resin composite are chipped during the finishing process, adding additional resin composite easily repairs these areas. The incisal and facial embrasures are opened with a thin separating disk (Fig 17-35m), the occlusion is adjusted, and the provisional restorations are smoothed and polished (Figs 17-35n to 17-35p). Finally, the provisional restorations are coated with an adhesive to add a shine. This can improve the patient's initial esthetic impression of provisionals (Fig 17-35q). The entire restoration is light cured for 30 seconds per tooth (Figs 17-35r and 17-35s).

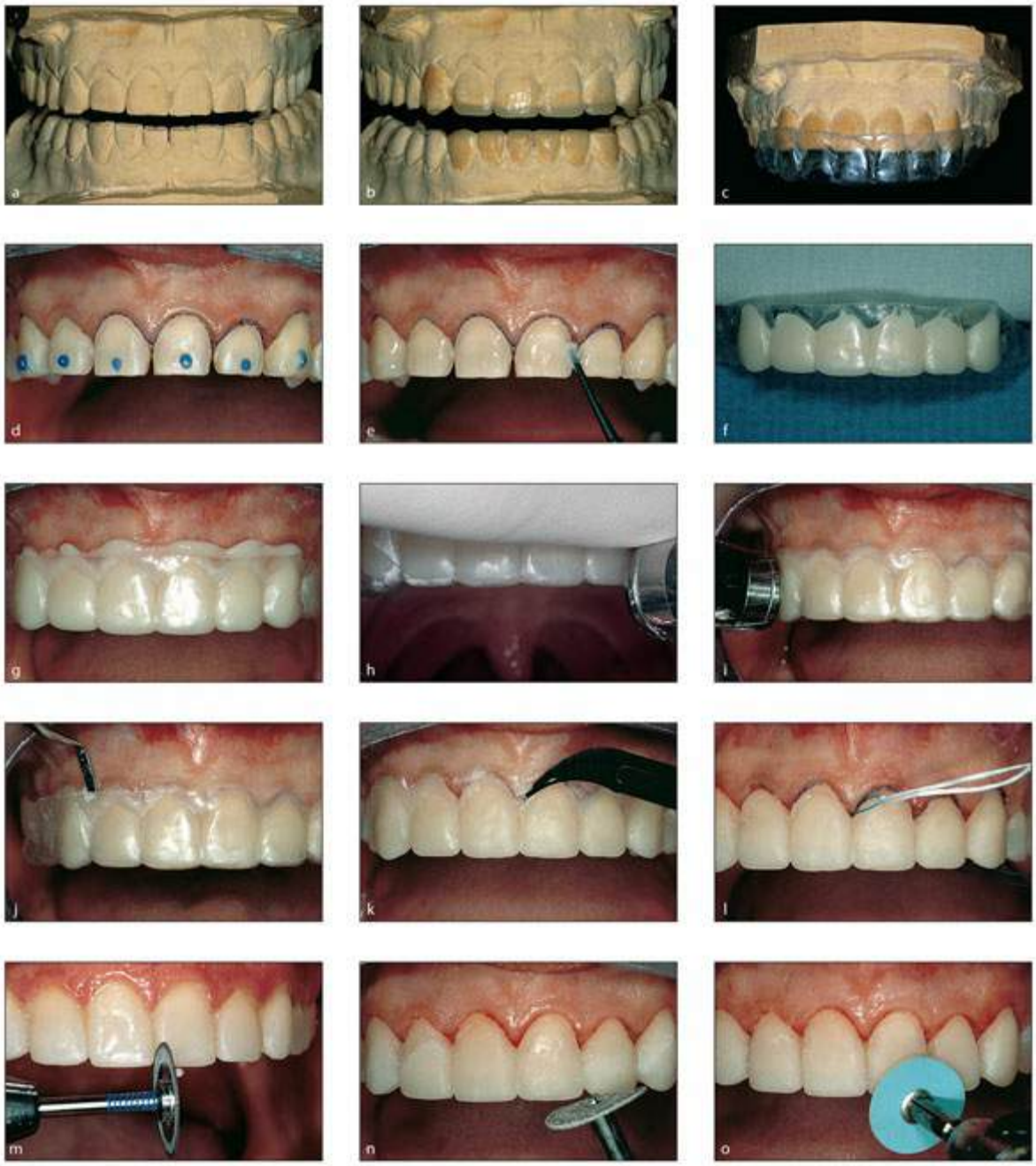


Fig 17-35 Fabrication of provisional restorations. *(a)* Preoperative casts. *(b)* Diagnostic wax-up. *(c)* Clear stent made from a duplicate cast of the diagnostic wax-up. *(d)* Prepared teeth are spot etched with 30% phosphoric acid on the incisal third. *(e)* The entire surface of each prepared tooth is coated with resin adhesive and light cured. *(f)* The facial and incisal areas of the clear matrix are filled with resin composite. *(g)* The filled matrix is placed over the prepared teeth. *(h)* The gingival two-thirds of the matrix is covered with a finger, while the incisal third is light cured for 10 seconds per tooth. *(i)* The gingival two-thirds is lightly cured for 0.5 to 1 second per tooth. *(j)* The matrix is gently teased away from the preparations to ensure that the resin composite does not stick to the matrix. *(k)* A no. 12 or 12B scalpel blade is used to remove the partially cured resin composite from proximal and lingual surfaces. *(l)* Floss and a floss threader are used in each gingival embrasure to ensure patency and overhang-free margins. *(m)* Facial and incisal embrasures are refined with a thin diamond disk. *(n and o)* A series of composite finishing disks are used to finish the facial surfaces.

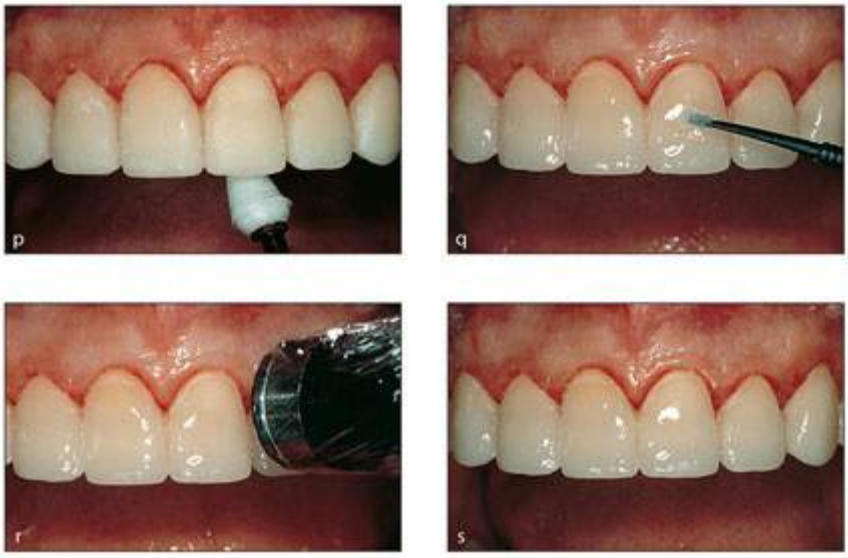


Fig 17-35*(cont) (p)* A composite polishing point is used to smooth and polish the lingual surfaces. *(q and r)* The facial surfaces of the provisional restorations are coated with an adhesive resin and light cured. *(s)* Completed provisional restorations.

Alternatively, the provisional restoration may be made in the laboratory. After the veneer preparations are completed, an impression is made and poured in fast-setting die stone (Snap-Stone, Whip Mix). The cast is separated from the impression in 5 minutes and covered with a separating medium; the provisional restoration is constructed with the same matrix technique as previously described. The provisional restoration, which is constructed from either bis-acryl or resin composite, can then be cemented with polycarboxylate cement or temporarily bonded with a resin composite as previously described (see Fig 17-5).

At the appointment for placement of the definitive veneers, the provisional restorations are removed. The resin composite over the small area of etched enamel in the incisal third of the facial surface is lightly removed with a diamond bur, cutting dry. If water is used, it is very difficult to determine the interface between the provisional composite and the tooth structure. The remaining resin composite is flicked off with a spoon excavator. If a veneer does not seat during the try-in, there is probably resin composite from the provisional restoration remaining in the etched area. The preparation should be closely inspected to ensure that all of the resin composite has been removed.

Placement

The anatomy of a porcelain veneer is illustrated in Fig 17-36. The laminate veneer

can be made from feldspathic porcelain, pressed/milled monolithic ceramic with stain, or pressed/milled core porcelain with a veneering ceramic addition. Advantages and disadvantages of each ceramic are discussed in chapter 18. The inner surface of the veneer must be etched with hydrofluoric acid or another ceramic etchant. This step is usually accomplished in the laboratory. The etching time provided by the manufacturer must be followed closely. In addition to the microporosities that assist in micromechanical retention, microcracks are increased as etching time increases. These microcracks decrease the flexural strength of the porcelain and weaken the veneer.[10] (For a detailed description of the steps in veneer placement, see Box 17-1.) The veneers are first tried in individually for marginal fit. They are then tried in together to ensure that interproximal contacts are correct. Finally, one veneer (or more) is filled with water or try-in paste and taken to the mouth for the color try-in. If the color is acceptable to the patient with water or water-soluble try-in paste, the dentist proceeds with the bonding procedure as outlined in Box 17-1.

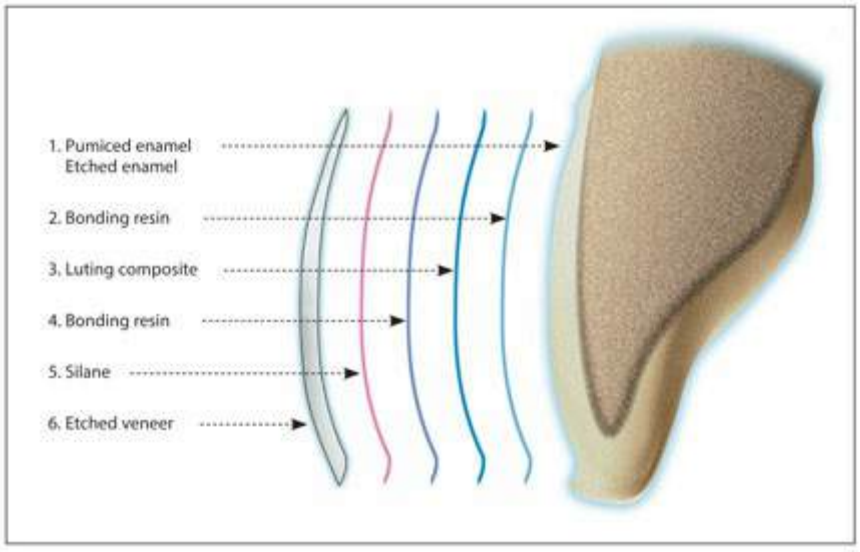


Fig 17-36 Anatomy of a porcelain veneer.

Box 17-1	**Procedures for insertion of porcelain veneers**
Veneer try-in 1. Check the veneers for fit on dies, and transilluminate to check for fracture lines. 2. Try in the veneers individually for fit and then all togeth er. Interproximal contact areas may need to be adjusted with a microfine diamond or disk. Do	

not make any other adjustments until the veneers are bonded.

3. Choose a shade of resin composite or water-soluble try in paste, place on the inside of the veneer, and try in.

4. If water-soluble try-in paste has been used, wash the veneer with water and air dry before loading with the unfilled resin and luting composite. If the shade is correct, skip steps 5 through 12, and proceed with step 13.

5. If the shade is incorrect, remove the try-in composite and select another shade, or customize the shade by adding tint to the luting composite.

6. If characterization (eg, blue in incisal areas, yellow at gingival areas) is required, tints and opaques should be placed only on the tooth and not on the inside of the veneer. The tints and opaques are brushed on the tooth in a thin layer and light cured for 30 seconds. All try-in composite must be removed from the tooth (facial, proximal, and lingual aspects) before the tints and opaques are cured. Because the tooth was not etched, the cured tints and opaques can be scraped off easily with an explorer at the end of the try-in.

7. Once a combination of composite, tint, and/or opaque has been determined, make note of it so that it can be reproduced exactly for final luting.

8. Clean the try-in resin composite from the inside of the veneer with acetone using two different beakers. Clean the bulk of the resin composite with a

14. Place clear plastic matrix or dead-soft metal matrix material on the distal aspects of both central incisors.
15. Etch both central incisors with 30% to 40% phosphoric acid for 20 seconds, wash for 3 seconds, and air dry to ensure adequate etch of the enamel.
16. Remoisten tooth surface with water or chlorhexidine.
17. If using a three-step etch-and-rinse bonding agent, actively apply several coats of primer and gently air dry until the tooth surface is completely dry. Check for a uniform shiny surface. If using a two-step etch-and-rinse or self-etching adhesive system, follow manufacturer instructions for drying and curing.
18. If using a three-step etch-and-rinse bonding agent, place the dentin adhesive on the tooth surface and the inner surface of the veneer. If using a two-step etch-and-rinse or self-etching adhesive system, place the component that contains the bonding resin on the inner surface of the veneer.
19. If tints or opaques are necessary, place them on the pre determined areas of the preparation and light cure for 90 seconds.

Placement

20. Place bonding resin and light-cured resin composite into the veneer being bonded. The operatory light should be turned off at this time.
21. With shimstock (0.0005-inch thickness) between the central incisors and clear plastic matrix strips or dead soft metal matrix material on the distal aspects of each

brush dipped in the first beaker, then transfer the veneer to the second, clean beaker of acetone to remove the remaining resin composite.

Veneer preparation

9. Place the veneers, etched side down, on a 2 × 2–inch gauze pad in a glass beaker of clean acetone, and place the beaker in an ultrasonic cleaner for 5 minutes.
10. Remove the veneers from the acetone and dry.
11. If the veneers have already been etched, place silane on the inner surface of the veneers and allow it to air dry. If not already etched at a dental laboratory, etch with hydrofluoric acid and then add silane as per instructions.

Tooth preparation

12. Place retraction cord in the sulcus of each prepared tooth (but not the mandibular incisors if the margins are more than 1.0 mm from the gingival crest).
13. Clean both central incisors with oil-free pumice paste. (This should be done for each tooth just before etching.)

Ensure that all excess resin composite is removed on the facial and proximal surfaces. Finishing of the mar gins with a rotary instrument should be avoided, because it will damage the root and remove the glaze from the veneer. This will cause an increase in plaque retention and elicit a gingival reaction.

30. Remove excess resin composite from the lingual sur faces with an egg-shaped, 12-fluted central incisor, gently place veneers onto both central incisors and tease into place. Ensure that excess resin composite appears at all margins.
22. Remove excess resin composite from veneers with a small brush or explorer, depending on viscosity.
23. While standing in front of the patient, visually inspect to ensure that veneers are placed correctly. Ensure that mesial surfaces are in contact.
24. Lightly press the facial surfaces of the veneers and light cure from the lingual aspect for 30 seconds, then light cure from the facial aspect for 30 seconds.
25. Remove the excess resin composite only on the distal surfaces of the central incisors with a no. 12 or 12B scalpel blade to ensure that the veneers on the lateral incisors will fit.
26. Visually inspect for voids and repair if possible.
27. Try in the left lateral incisor veneer to ensure correct fit.
28. Repeat steps 14 to 28 for each veneer being placed.

Finishing

29. Remove minimal gingival flash of resin composite with a no. 12 or 12B scalpel blade. Move blade from the veneer gingivally to avoid chipping the veneer's margin. with a finishing diamond or bur. Be careful not to scar the cementum.
36. Polish all roughened porcelain using an abrasive impregnated rubber porcelain polishing system (Bras seler or Shofu). 37. Finish the proximal areas with finishing and polishing strips. 38. Any visible porcelain that has been finished and smoothed with a rubber point should be polished with diamond polishing paste on a wet felt wheel and pro phylaxis cup at the gingival margin. Be careful not to polish the cementum. 39. With the retraction cord still in place, re-etch and rebond (with rebonding or surface sealing resin) all margins.
40. Remove the retraction cord.
41. Have the patient return in 1 week to inspect for excess composite and rough areas. At this time, final esthetic reshaping may be accomplished.

carbide bur. 31. Smooth lingual surfaces with composite polishing points or disks. 32. Check and adjust the occlusion in maximum intercus pation position and in excursion (with special attention to the distal incisal edges of the maxillary lateral inci sors). 33. Reshape incisal edges and contours while standing in front of the patient. 34. Reshape and contour incisal embrasures with a finish ing diamond or thin separating disk. 35. Gingival margins should be smooth and require no finishing. If gingival margins are rough, smooth them	

There are many resin luting cement kits available with differing degrees of translucency and viscosity (Fig 17-37). Translucent cements are indicated as the standard material for veneer bonding. The more opaque cements tend to block the natural tooth color, resulting in a veneer with a less-natural appearance. The opaque cements are more commonly used to help block the darkness of discolored teeth. However, the use of these cements can result in a monochromatic appearance and an opaque line at the thin gingival margin of the veneer (Fig 17-38). It is preferable to block the darkness of discolored teeth with a layer of masking dentin porcelain, and/or with porcelain modifiers, in the body of the veneer rather than with an opaque cement (Fig 17-39).

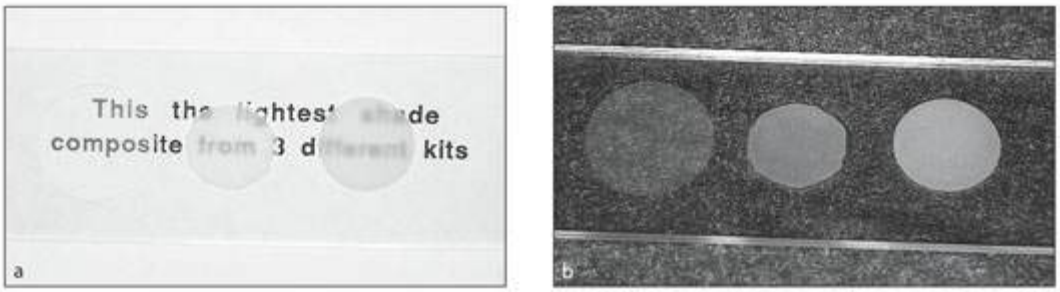


Fig 17-37*(a and b)* The lightest shade of luting composite from three different kits has been placed between two glass slides. The same materials are shown against two different backgrounds. Note the differences in translucency. (Fig 17-37a reprinted from Robbins[49] with permission.)



Fig 17-38*(a)* Moderately dark, tetracycline-stained teeth prior to veneer preparation. *(b)* Veneers bonded on dark teeth with opaque resin cement. Note the opaque gingival margins.



Fig 17-39 Veneers displaying different amounts of translucency. The left veneer is made from a very translucent porcelain, which allows most of the underlying tooth color to show through. The center veneer has a base layer of masking dentin porcelain, which is used to block dark underlying tooth color yet maintain some degree of polychromicity. The right veneer has a layer of opaque resin cement bonded to the inner surface of the veneer. The opaque resin is very reflective and results in a displeasing, monochromatic appearance.

The second major difference in veneer luting cements is their viscosity. Initially, most resin cements had a low viscosity so that the veneers could be placed with minimal pressure, thereby decreasing the risk of fracture.[50] However, the low-viscosity resins have some disadvantages: *(1)* Because of their honeylike consistency, it is more difficult to ensure correct veneer placement, especially when the veneers have no positive stop (ie, no incisal overlap); *(2)* cleanup of excess resin is more difficult; and *(3)* at least theoretically, the physical and mechanical properties are compromised because of the increased proportion of resin matrix. The major indication for the low-viscosity luting resin is the all-ceramic crown or a veneer that covers most of the tooth surfaces. Because of friction, high-viscosity luting resins will not allow complete seating of these restorations. Recently, interest in the higher-viscosity luting resins has increased because they overcome most of the disadvantages of the low-viscosity luting resins.

Friedman[51] has described a technique for using the highly filled resin composite from a standard restorative resin composite kit. The material is brought to room temperature and placed into the veneer in a thin layer through a ribbon tip. The thixotropic properties of the resin composite allow the highly filled material to flow

under moderate seating pressure. However, placing the ampule of composite in a hot water bath (160°F) or composite warmer (eg, Calset, AdDent) before ejecting the composite is a more effective method of improving the flow characteristics of the resin composite. With this technique, the seating of the veneers can be more accurately controlled, and cleanup is simplified (see Box 17-1).

The third difference between luting agents is in the chemical mechanism by which curing is initiated. The preferred method for bonding porcelain veneers is a light-curing luting composite. Light curing allows a longer working time, shorter finishing time, and superior color stability compared with dual-curing or chemically cured material. It used to be that if the porcelain thickness was greater than 0.7 mm, the light-curing composites did not reach their maximum hardness.[52] However, with current light-curing units, it is possible to cure through 2 mm or even greater thicknesses of ceramics.[53–55] It is advisable to avoid using a dual-curing resin luting cement because light-curing resin composites tend to be more color stable than dual-curing systems.[56] However, the ability of the curing light to penetrate through ceramic to the underlying adhesive and resin cement is dependent on the ceramic thickness, shade and translucency.[57]

In those situations in which the dentist is concerned about the adequacy of cure of a light-curing resin cement through a veneer, a dual-curing composite system is recommended. Dual-curing materials contain the initiation systems for both chemical-curing and light-curing composites. Even so, dual-curing resin materials should always be light cured, because any light will enhance the degree of conversion of a dual-curing material[58] and will provide for better color stability.[59] Additionally, if the veneer is to be bonded to a tooth preparation that includes a significant amount of dentin, light curing of the adhesive separately may produce significantly higher bond strengths to dentin than the traditional co-curing method (ie, not light curing until after the resin cement application).[60] However, curing the adhesive prior to luting the veneer runs the risk that cured, pooled adhesive will prevent adequate adaptation of the veneer to the preparation.

Color Management and Characterization

A common problem with porcelain veneers is the lack of color differentiation between the gingival and incisal portions of the restoration. Several methods of color characterization can be used to correct the monochromatism.[49,61,62] The best and most basic method makes the color changes in the porcelain itself. A color

diagram that outlines the desired shade and color changes and any other special characterization, such as hypocalcified or hyperchromatic areas, can be given to the laboratory technician. However, the most effective method of communicating color to the laboratory technician is with photographs (see Figs 17-8b and 17-8c). When characterization is incorporated into the veneer, it is permanent. If the esthetic result is not satisfactory during the try-in, it is difficult, if not impossible, to successfully modify the veneer.

A second commonly used method involves the modification of the color with the underlying luting composite. All porcelain veneer kits have several different shades of luting composite. If an appropriate shade of luting cement is not available, resin tints can be added to the luting composite to effect virtually any desired color.

A third and less commonly used method of characterization involves the direct placement of the resin tints on the tooth before placement of the veneer (Fig 17-40). During the try-in, the desired resin tints are placed on the tooth and light cured. The chosen base shade of luting composite is placed in the veneer, and the veneer is placed on the tooth. A determination is then made regarding the esthetic result. The veneer can be removed, and the cured resin tint on the tooth can easily be scraped off with an explorer because the enamel was not etched. If the esthetic result is acceptable, the dentist can proceed with the bonding, using the tints that were used during the try-in.



Fig 17-40 *(a)* Veneer preparations on the maxillary central and lateral incisors. *(b)* Placement of yellow resin tint in the gingival third and blue resin tint in the incisal third. *(c)* Kit of resin tints and opaques. *(d)* Monochromatic appearance of veneers tried in with try-in paste but without the use of resin tints. *(e)* Polychromatic appearance of bonded veneers after the placement of resin tints. (Reprinted from Robbins[62] with permission.)

The definitive restoration will not always exhibit the same color displayed at the

try-in. This occurs because the luting composite becomes more translucent when it is cured, allowing more of the underlying tooth color to show through.[63] Although not usually a problem, this color-change phenomenon will occasionally result in a disappointing esthetic result, especially when the dentist is attempting to match a single veneer to an adjacent natural tooth.

Discolored teeth

Darkly discolored teeth present the greatest challenge for porcelain veneers because the underlying tooth color, cement color, and ceramic thickness all influence the resulting color of the veneer.[64] There are many causes of tooth discoloration, including extrinsic staining, fluorosis, pulpal injury, drugs (eg, tetracycline), and previous restorations. The ideal method of dealing with stain is to remove it, when possible. Extrinsic stains are easily removed during tooth preparation. Because fluoride predominantly affects the enamel, the discoloration of fluorosis is also commonly diminished by tooth preparation.

However, the by-products of pulpal injury, tetracycline, and previous restorations are found predominantly in the dentin, making their removal more difficult. The restoration of the high-chroma, low-value tetracycline-stained tooth with a porcelain veneer is perhaps the most difficult treatment situation. Instinctively, practitioners prepare the discolored teeth more deeply to allow more room for porcelain. However, as more enamel is removed, the underlying color becomes darker (Fig 17-41). Herein lies the major difficulty in placing porcelain veneers on tetracycline-stained teeth. Additional tooth structure should only be removed in the areas of discoloration, not over the entire preparation. Tetracycline-stained teeth are often severely discolored in the middle third of the tooth, with other areas having an acceptable or easily modified color. Ideally the preparation should stay within the enamel in the gingival and incisal thirds and only deepen in the middle third. The amount of reduction required depends on the color of the preparation and the desired chroma/value of the definitive restoration.



Fig 17-41 Preparation of a severely tetracycline-stained maxillary central incisor, which demonstrates increased darkness with increased depth of preparation.

Ceramic buildup

The evolution of contemporary porcelain materials and sophisticated fabrication techniques has allowed laboratory technicians to block out significant discolorations with ultrathin veneers (Fig 17-42). Veneers with pressed or milled ceramic cores provide the technician the ability to block out many of the discolorations without overly opaquing the veneer and destroying the natural beauty of the restoration.[64] The color of the prepared tooth is used for the development of natural color. It is important that clinicians provide the laboratory with a photograph of the final preparation at a 1:1 magnification. With this photograph, the tooth can be measured by the technician and the maverick colors or discolorations located. Once blemishes have been identified, masking agents can be added selectively to the restoration. Color characteristic stains are routinely used to create optical effects and conceal discolorations in conservative veneer restorations. These stains are opaque metal oxides that are painted into wet porcelains. The stains are used to mask discoloration or further characterize the appearance of the veneer and are subsequently covered with a thin layer of opalescent or fluorescent porcelain prior to firing. It is possible to modify the color of the tooth by one to two shades for every 0.2 mm of tooth reduction.



Fig 17-42 *(a)* Removal of bonded resin composite veneers demonstrates significant discoloration of the maxillary right canine and left central incisor. *(b)* Final conservative full-veneer preparations. *(c)* Porcelain veneers bonded with translucent resin demonstrating significant color correction with contemporary porcelains.

Masking

It is difficult to mask dark underlying tooth color with a natural-appearing porcelain veneer (Fig 17-43). If the luting cement is sufficiently opaque to mask the dark color, the definitive restoration commonly appears lifeless and monochromatic.[65] Conversely, if the veneer and/or cement contains no opaque element, the darkness of the underlying tooth structure will be visible through the porcelain. As previously stated, it is preferable to incorporate the opaque elements into the porcelain rather than to attempt to mask the darkness with opaque resin composite. This requires excellent communication between the laboratory technician and the dentist.

Postpreparation photographs with a shade tab for color comparison are very helpful to the technician when porcelain veneers are being fabricated for discolored teeth (see Fig 17-21c). The photographs illustrate both the intensity of the darkness and the position of the bands of darkness.



Fig 17-43 *(a)* Discolored maxillary anterior teeth prepared for porcelain veneers. *(b)* Discolored maxillary anterior teeth with bonded porcelain veneers. (Reprinted from Robbins[62] with permission.)

Several methods of masking dark underlying tooth color with resin composite have been proposed. Friedman[51] reported successful use of a standard highly filled restorative resin composite as the luting cement. Reid[61] discussed the use of complementary colors to mask underlying dark tooth color. For example, if the tooth is predominantly yellow, a violet tint is placed and light cured. This neutralizes the high-chroma yellow to a low-value gray. Opaque resin tint is then placed over the violet tint and light cured to increase the value, and the veneer is bonded in the routine manner. Although theoretically appealing, this technique has not proven to be of practical benefit.

When discoloration is localized rather than generalized, it can be removed mechanically with a bur and replaced with the appropriate shade of glass ionomer or resin composite. This is usually done after the veneer preparation is completed but before the final impression is made; however, it can be done immediately before cementation of the veneer.

Composite luting agent can be used to block underlying darkness to a limited degree. However, most of the masking must be accomplished with body modifiers that are incorporated into the porcelain veneer. Depending on the degree of darkness being masked, the technician must place 50 to 100 μm of die spacer on the refractory die prior to veneer fabrication. This allows space for an additional thickness of the luting composite, which aids in the masking of the underlying darkness. Although the appearance of discolored teeth can usually be improved with porcelain veneers, the patient must understand the limitations of the restoration.

Bleaching

When a discolored tooth can be lightened before veneer placement, the final result is routinely improved. This can be accomplished in endodontically treated teeth with a "walking bleach" technique[66] (Fig 17-44) (see also chapter 16). A 2.0-mm-thick base of glass-ionomer cement is placed in the base of the pulp chamber to protect the cervix from bleach. A paste mixture of sodium perborate and water is placed in the pulp chamber, which is sealed with a temporary cement for 3 to 5 days. This can be repeated for several treatments until the desired result is attained. The mixture is ultimately removed, and the access area is restored with a resin composite restoration. At this point, the tooth is prepared for the porcelain veneer restoration. If the darkness returns in future years, the walking bleach can be accomplished again through the lingual access without disturbing the porcelain veneer restoration.



Fig 17-44 *(a)* Discolored endodontically treated maxillary right central incisor. *(b)* Maxillary right central incisor after completion of walking bleach procedure. *(c)* Maxillary right central incisor with a porcelain veneer bonded in place. (Reprinted from Robbins[49] with permission.) (Porcelain veneer created by Danny Diebel, CDT, Austin, Texas.)

The success of bleaching vital teeth before veneer placement is not as clear-cut. It is known that vital teeth that have been bleached have the potential to revert toward their original color with time.[67] However, the effect that the placement of porcelain veneers has on this relapse is not known. If color relapse does occur, the veneer restorations will also get darker. There has been reported success in bleaching teeth that have veneers bonded to them (see chapter 16).

Bleaching teeth with 35% hydrogen peroxide[68] or carbamide peroxide[69] immediately before the bonding procedure has a catastrophic effect on the resultant bond strength. Any bonding procedure should be delayed at least 1 to 2 weeks after the completion of bleaching to maximize bond strength and allow for shade stabilization. It is hoped that future research will clarify the effects of bleaching on the success of porcelain veneers.

Crown and Veneer Combinations

When a combination of veneers and crowns is placed, all restorations must be tried in individually and then simultaneously for fit. They must also be tried in for color evaluation with a try-in medium. If the veneer and crown shades do not match, it is preferable to make minor color modifications with the veneer luting resin to see if the veneers can be made to match the shade of the crowns, because these modifications can be made chairside. If that is not successful in matching the shade of the veneers and crowns, it is possible to attempt surface staining of the crowns. However, unless the office has a glazing oven, the crowns will have to be returned to the dental laboratory to complete the staining process. If it is necessary to return the case to the laboratory, then the dentist has the option of luting in whichever restorations the patient prefers, obtaining new shade photographs, and having the laboratory adjust the shade of the remaining restorations to the newly luted restorations. Alternatively, the entire case can be returned for refabrication and better shade matching.

Failure Repair

As mentioned earlier in the chapter, there are a number of causes for veneers to fail. However, it is possible to repair many such failures. When the luting composite remains on the inner surface of the veneer after a debonding event, it must be removed before the veneer can be rebonded. The bonding resin is removed either with air abrasion or a fine finishing diamond bur. The veneer is then cleaned with acetone and re-etched with 9.5% hydrofluoric acid for 20 to 90 seconds, depending on the type of porcelain. If 9.5% hydrofluoric acid is not available, 1.23% acidulated phosphate fluoride can be used to etch the porcelain; however, this requires a 10-minute etching time. The veneer is then washed, dried, silanated,[70] and rebonded. The patient must know that there is a significant risk of veneer fracture during removal of the luting composite.

A small percentage of veneers will fracture.[3–5] It is possible to repair fractured porcelain. First, the porcelain fracture site is etched with 9.5% hydrofluoric acid for 20 to 90 seconds. After the veneer is washed and dried, silane is placed and allowed to air dry. The repair is then accomplished with conventional bonded resin composite techniques. Because the hydrofluoric acid should not be allowed to contact natural tooth structure or soft tissue, this etching procedure should be performed with rubber dam isolation.

As noted, marginal staining around the veneer can occur. If the marginal stain is

superficial, it can be removed by tray bleaching with 10% carbamide peroxide for several days. After the stain has been removed, the margin can be etched with 30% to 40% phosphoric acid and sealed with a dentin bonding agent. If the stain is slightly penetrating at the margin (Fig 17-45), it can be mechanically removed with a small bur and the area restored with conventional bonded resin composite. When there is significant penetration of stain under the veneer, the entire veneer must be removed (Fig 17-46).



Fig 17-45 Minimally penetrating stain at the mesi-al margin of the maxillary right canine.



Fig 17-46 Deeply penetrating stain under the porcelain veneer on the maxillary right central incisor.

Veneer Removal

Removal of porcelain veneers is not only time-consuming but also difficult and technique sensitive, especially if the underlying tooth color is light. The veneer cannot be grooved with a bur and rotated (torqued) in the same manner that a cemented gold crown is removed. The veneer must be removed with a diamond bur in the same way that enamel is removed during initial tooth preparation.

The dentist starts removing the porcelain in the midfacial area with a back-and-forth sweeping motion with a barrel diamond (Fig 17-47a). This must be done without water spray so that the operator can visualize the subtle color difference between the veneer and tooth structure as the interface is reached. Therefore, the dental assistant must cool the tooth with a constant stream of air. Once this interface is apparent (Fig 17-47b), the diamond bur is moved laterally away from the area of exposed enamel toward the periphery of the preparation. Care must be taken to

remove as little enamel as possible during this step.



Fig 17-47 Veneer removal. *(a)* Initial removal of porcelain before tooth structure is reached. *(b)* Porcelain is removed until the first area of enamel is visualized. *(c)* Exploring margins with a no. 12 or 12B scalpel blade to remove remaining resin composite and porcelain.

The procedure continues until only a small amount of porcelain remains at the margins. If there has been microleakage at the margins, the remaining marginal porcelain can be removed with a no. 12 or 12B scalpel blade. However, if the marginal seal is intact, the remainder of the porcelain must be cautiously removed with the diamond bur.

It is very important that the operator not lose orientation in relation to the porcelain-tooth interface. If this occurs, it is very easy to inadvertently remove all of the enamel. For this reason, one veneer should be completely removed before starting veneer removal on the next tooth.

After all porcelain veneers have been removed, gingival retraction cord is packed to clearly expose all margins. As the final step of veneer removal, all margins are explored with a sharp no. 12 or 12B scalpel blade (Fig 17-47c), which commonly results in the removal of additional small areas of residual resin composite and porcelain.

Maintenance

The maintenance of the porcelain veneer restoration is similar to that of the porcelain crown. Devices such as an ultrasonic cleaner, air-abrasive polisher, and prophylaxis cup with pumice must be avoided. Surface stains may be removed from porcelain veneers with aluminum oxide polishing paste or diamond polishing paste on a felt wheel or rubber cup. Proximal stains may be removed with composite polishing strips. When scaling is performed around porcelain veneers, care must be taken not to chip the margins. If a fluoride preparation is needed by the patient, it should have a neutral pH; because of their ability to etch porcelain, neither acidulated phosphate fluoride nor stannous fluoride should be used.

The patient should be advised that foods and liquids with a high potential for

staining, such as coffee and tea, increase the potential for marginal staining. The patient also must be made aware of the potential for the porcelain to fracture. Activities such as ice chewing and fingernail biting must absolutely be avoided. It is a good idea to make an occlusal guard appliance for all patients who have porcelain veneer restorations. When porcelain veneer restorations will oppose natural teeth or when the patient has a history of a parafunctional habit, a protective appliance should be fabricated to protect both the porcelain veneers and the opposing teeth.

Acknowledgment

The illustrations for chapter 17 were completed by John Bonfardeci of Studio Giovanni.

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